

SMS SPAM DETECTION USING DYNAMIC EVIDENCE ROUTING TRANSFORMER

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Abstract—Spam SMS messages have become a major challenge due to their use in phishing attacks, financial fraud, malicious links, and unsolicited advertisements. Traditional spam detection methods often rely on fixed representations and may not capture the different forms of evidence present in short messages. This paper presents a Dynamic Evidence Routing Transformer for SMS spam detection. The system uses a Transformer-based encoder with three specialized expert branches: Semantic Expert, Pattern Expert, and Context Expert. A Dynamic Evidence Router learns instance-specific weights for each expert and performs dynamic weighted evidence fusion before the final SPAM/HAM classification. The SMS data are preprocessed and divided into training, validation, and test sets. The current implementation is evaluated using Accuracy, Precision, Recall, and F1-Score, with additional ablation and robustness experiments defined for component analysis. On the current held-out test set of 517 messages, the implemented model achieved 99.42% Accuracy, 100.00% Precision, 95.38% Recall, and 97.64% F1-Score. The approach provides an adaptive way to combine semantic, local-pattern, and contextual evidence for SMS spam detection.

Keywords—SMS spam detection, Transformer, DistilBERT, dynamic routing, evidence fusion, text classification, natural language processing.

I. INTRODUCTION

SMS communication remains an important channel for personal, commercial, and service-related communication. However, unwanted messages can be used for advertisements, fraudulent offers, phishing attempts, and malicious links. Because SMS messages are short, their meaning may depend on a small number of highly informative words, local token patterns, and contextual relationships. This makes spam detection a challenging text-classification problem.

Earlier SMS spam filtering research established the value of content-based classification and public benchmark datasets for comparing spam filters [1], [5]. Conventional machine learning methods can perform well when discriminative lexical features are available, but fixed feature representations do not explicitly adapt the relative importance of different evidence types for every message.

Transformer language models provide contextual representations by processing relationships between tokens. BERT introduced bidirectional Transformer representations for language understanding [2], while DistilBERT provides a smaller distilled version intended to retain much of the language understanding capability of BERT with lower computational cost [3]. These properties make Transformer encoders useful for short-text classification.

This work proposes a Dynamic Evidence Routing Transformer for SMS spam detection. The proposed architecture uses a DistilBERT encoder followed by three expert branches: Semantic Expert, Pattern Expert, and Context Expert. Instead of combining these branches with fixed weights, a Dynamic Evidence Router learns instance-specific weights for each expert. The weighted representations are then combined through Dynamic Weighted Evidence Fusion and passed to a classification layer. The objective is to allow the model to emphasize the evidence that is most useful for the particular SMS being analyzed.

II. RELATED WORK

SMS spam filtering has been studied using content-based methods, feature engineering, and machine learning classifiers. The SMS Spam Collection introduced by Almeida, Gómez Hidalgo, and Yamakami provides a public corpus of labeled SMS messages for spam-filtering research and comparison [1]. Earlier work also investigated feature engineering for mobile SMS spam filtering and demonstrated the importance of carefully selected textual characteristics [5].

The introduction of BERT demonstrated that pretrained bidirectional Transformer representations can be fine-tuned for downstream natural language understanding tasks [2]. DistilBERT was subsequently proposed as a smaller and faster alternative based on knowledge distillation [3]. These models are useful for text classification because they provide context-aware token representations without requiring manually designed linguistic features.

Dynamic routing was introduced in neural architectures as a mechanism for adaptively directing lower-level outputs toward appropriate higher-level representations [4]. Inspired by the general idea of adaptive routing, the proposed work uses a different formulation: the router predicts instance-specific weights over the outputs of semantic, pattern, and context experts. Thus, the proposed routing mechanism is designed specifically for evidence selection in SMS classification rather than capsule routing.

III. DATASET AND DATA PREPROCESSING

The project uses the SMS Spam Collection, a public dataset containing SMS messages labeled as ham or spam [1]. The UCI record for the dataset reports 5,574 text instances and identifies the task as classification [6]. The original collection contains legitimate and spam messages collected from multiple sources.

For the current implementation, the SMS records are cleaned and prepared for Transformer-based processing. Invalid or duplicate records are handled during preprocessing, labels are encoded into the two target classes, and messages are tokenized for the DistilBERT encoder. The processed data are divided into training, validation, and test subsets.

The current implementation used 4,139 training messages, 517 validation messages, and 517 test messages. The held-out test set therefore contains 517 messages. The training set is used for parameter learning, the validation set supports model selection and monitoring, and the test set is reserved for final performance measurement.

IV. PROPOSED DYNAMIC EVIDENCE ROUTING ARCHITECTURE

The proposed architecture is designed around the observation that different SMS messages may contain different useful forms of evidence. A promotional message may contain strong local patterns, while a conversational message may require more semantic or contextual information. The architecture therefore separates evidence extraction from evidence weighting.

A. DistilBERT Encoder

The input SMS is tokenized and passed through a pretrained DistilBERT encoder. The encoder generates contextual representations of the message. DistilBERT is used as the backbone because it provides a compact Transformer-based representation suitable for text classification [3].

B. Semantic Expert

The Semantic Expert focuses on the overall meaning of the message. It receives the contextual representation produced by the encoder and transforms it into a representation intended to preserve message-level semantic information.

C. Pattern Expert

The Pattern Expert focuses on local textual patterns that may be useful for spam detection. It processes the encoder representation to emphasize short-range or token-level patterns such as characteristic wording, repeated structures, or promotional expressions.

D. Context Expert

The Context Expert focuses on informative contextual relationships within the message. It does not perform explicit named-entity extraction. Instead, it transforms the Transformer representation to emphasize contextual information that can help distinguish spam from legitimate messages.

E. Dynamic Evidence Router

The three expert representations are provided to a Dynamic Evidence Router. The router learns instance-specific weights

for each expert rather than applying one fixed weighting scheme to all messages. The learned weights determine how strongly each evidence source contributes to the final representation.

F. Dynamic Weighted Evidence Fusion

The weighted expert representations are combined using Dynamic Weighted Evidence Fusion. This produces a single classification representation that reflects the evidence selected by the router for the current SMS.

G. Classification Layer

The fused representation is passed to a classification layer that predicts one of two classes: SPAM or HAM. The complete inference flow is therefore: SMS input → preprocessing → DistilBERT encoder → three experts → dynamic evidence router → dynamic weighted evidence fusion → classification → SPAM/HAM.

V. SYSTEM IMPLEMENTATION

The implementation follows two logically separate stages: model development and inference. During model development, the dataset is preprocessed and divided into training, validation, and test subsets. The DistilBERT encoder and the three expert branches are trained together with the dynamic routing module. The trained model is saved for later inference.

During inference, a new SMS is first preprocessed and encoded. The trained DistilBERT model generates a contextual representation, which is passed to the Semantic, Pattern, and Context experts. The Dynamic Evidence Router produces instance-specific weights for these expert outputs. Dynamic Weighted Evidence Fusion combines the weighted representations, and the classification layer produces the final SPAM or HAM prediction.

The system is designed so that the training split and validation split are not part of the prediction flow. They belong to model development, whereas a user prediction uses the already trained model.

VI. EXPERIMENTAL SETUP AND EVALUATION

The current implementation evaluates the classifier using Accuracy, Precision, Recall, and F1-Score. Accuracy measures the proportion of correctly classified messages. Precision measures how many messages predicted as spam are actually spam. Recall measures how many actual spam messages are detected. F1-Score provides the harmonic mean of Precision and Recall.

In addition to the standard metrics, the project defines two further analyses. First, ablation testing removes or modifies individual expert components to examine their contribution to the final system. Planned variants include removing the Semantic Expert, Pattern Expert, or Context Expert, and replacing the dynamic router with a static fusion strategy. Second, robustness testing examines performance under challenging variations of SMS text. These experiments should be reported only after their corresponding runs are completed.

VII. RESULTS AND DISCUSSION

The current saved model was evaluated on a held-out test set of 517 SMS messages. The obtained results are shown in Table I. The model achieved 99.42% accuracy, with 100.00% precision, 95.38% recall, and 97.64% F1-score.

The confusion matrix contains 452 correctly classified HAM messages, 62 correctly classified SPAM messages, 3 SPAM messages classified as HAM, and 0 HAM messages classified as SPAM. These values correspond to 514 correct predictions out of 517 test messages.

The current routing analysis produced average expert weights of approximately 14.38% for the Semantic Expert, 82.60% for the Pattern Expert, and 3.02% for the Context Expert. These values are descriptive averages rather than a standalone measure of causal component importance. The planned ablation experiments are therefore necessary to determine how much each expert contributes to overall performance.

VIII. ABLATION AND ROBUSTNESS ANALYSIS

Ablation analysis is intended to test whether the performance of the full architecture depends on its individual components. The full model will be compared with versions that remove one expert at a time. A further comparison will replace the Dynamic Evidence Router with static fusion. The same test protocol and evaluation metrics should be used for each variant so that the effect of each architectural component can be measured consistently.

Robustness analysis is intended to examine whether the classifier remains reliable when SMS messages contain challenging textual variations. Examples can include altered spacing, unusual punctuation, spelling variations, repeated characters, or other realistic modifications. The robustness results should be reported separately from the current base-model results after the experiments are executed.

IX. APPLICATIONS AND ADVANTAGES

The proposed system can be used as a component of SMS filtering applications, messaging-security tools, and mobile communication systems. Its Transformer-based representation can capture contextual information, while the expert structure provides multiple evidence pathways for classification.

A key architectural advantage is instance-dependent evidence weighting. Instead of assuming that semantic, pattern, and contextual evidence are equally useful for every SMS, the router can assign different weights to the experts for different inputs. The approach also provides a modular structure in which individual experts can be analyzed through ablation experiments.

X. LIMITATIONS AND FUTURE WORK

The current study has several limitations. The SMS Spam Collection is an established benchmark but represents a particular collection of messages and may not fully represent modern spam campaigns. The current model also operates on message text and does not use sender identity, delivery metadata, or external web information. Therefore, performance on contemporary SMS threats should be validated using additional datasets.

Future work can include cross-dataset evaluation, larger and more diverse SMS collections, multilingual SMS detection, adversarial and robustness testing, improved routing strategies, and lightweight deployment. Further analysis of the instance-specific routing weights can also be used to study how the model changes its evidence selection across different message types.

XI. CONCLUSION

This paper presented a Dynamic Evidence Routing Transformer for SMS spam detection. The proposed system combines a DistilBERT encoder with Semantic, Pattern, and Context experts and uses a Dynamic Evidence Router to learn instance-specific weights for each expert. Dynamic Weighted Evidence Fusion then combines the selected evidence before SPAM/HAM classification.

On the current held-out test set of 517 messages, the implementation achieved 99.42% Accuracy, 100.00% Precision, 95.38% Recall, and 97.64% F1-Score. The current results demonstrate the feasibility of the proposed architecture, while ablation and robustness experiments remain important for determining component contributions and generalization.

TABLE I. PERFORMANCE OF THE CURRENT MODEL

Metric	Result
Accuracy	99.42%
Precision	100.00%
Recall	95.38%
F1-Score	97.64%
Test Set	517 SMS messages

TABLE II. CONFUSION MATRIX OF THE CURRENT TEST RUN

Actual / Predicted	HAM	SPAM
HAM	452	0
SPAM	3	62

TABLE III. AVERAGE ROUTING WEIGHTS IN THE CURRENT TEST RUN

Expert	Average Weight
Semantic Expert	14.38%
Pattern Expert	82.60%
Context Expert	3.02%



Fig. 1. Proposed training, inference, and offline evaluation workflow.

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